



Hardware Accelerator for Object Detection on FPGA

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Methodology: Development



Project Description

- Deep neural network enable real-time object detection in edge applications such as robotics, drones, and autonomous systems.
- FPGAs provide parallel hardware acceleration for these workloads while operating under limited embedded system resources.
- This project will develop a object detection accelerator on a Zynq based FPGA to reduce hardware cost while maintaining detection accuracy.



Project Deliverables

- Functional software baseline and quantized object detection model.
- Implemented FPGA hardware accelerator integrated with the Zynq platform.
- Performance comparison using accuracy, latency, memory bandwidth, and the FPGA resource utilization.



Project Dependencies and Concerns

- Access to the Zynq FPGA platform, Vivado/Vitis tools, Object detection models, and datasets.
- Limited digital signal processing, BRAM, logic, and memory bandwidth may restrict accelerator size and performance.
- Reduced numerical precision may lower detection accuracy, creating a tradeoff between hardware efficiency and model performance.